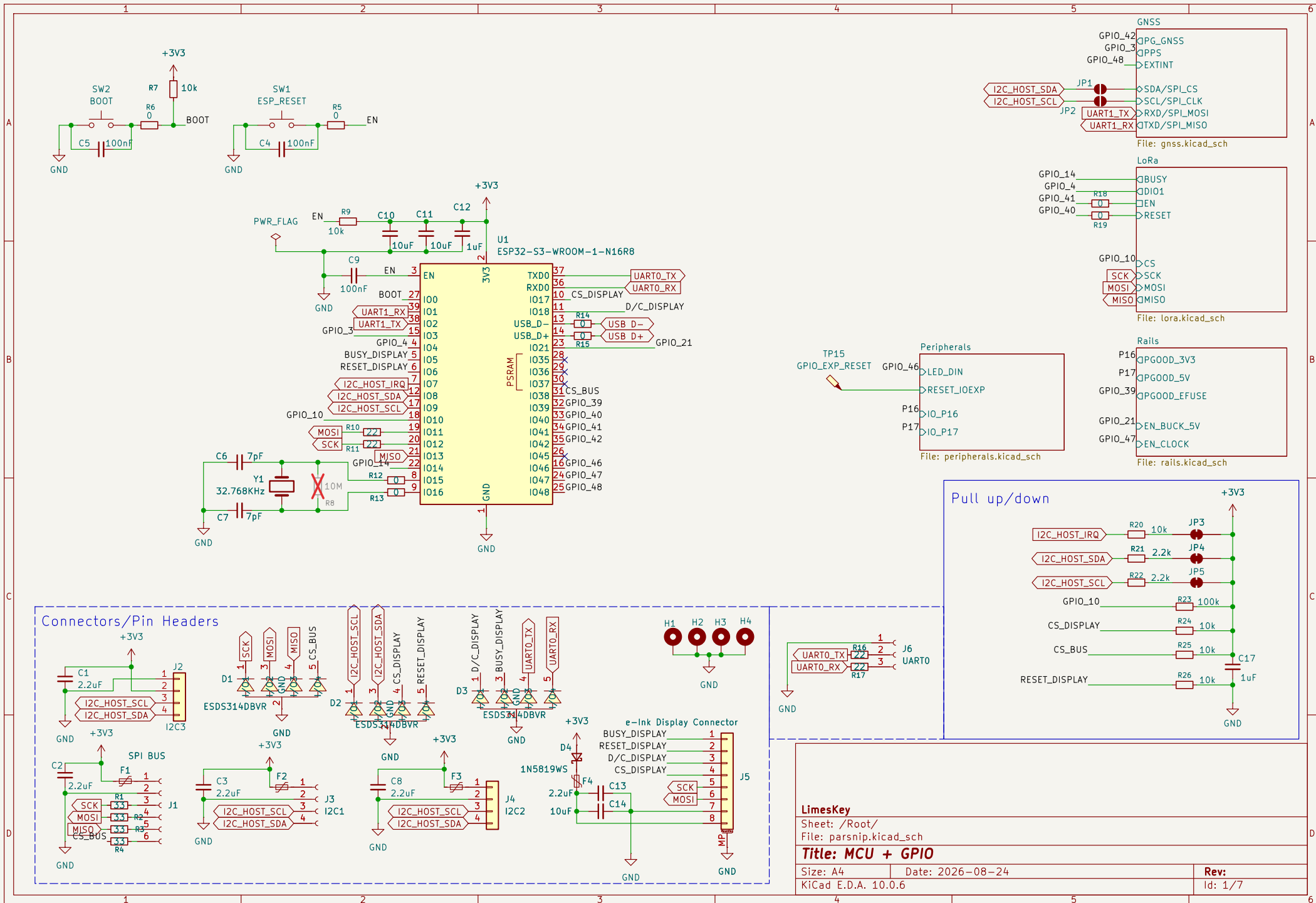




3V3 Buck

The schematic shows a 3V3 Buck converter using the LM61460AASQRJRRQ1 IC. The input is connected to VSYS through capacitors C122, C123, and C124. The feedback network consists of R104, R105, R109, and C125. The output is connected to +3V3 through capacitors C130, C132, C133, C137, C141, C143, C148, and C153. A bootstrap network (C131, L4, R112, R115, R116) is used for the CBOOT pin. The PWR_FLAG output is connected to the PWR_FLAG pin. Test points TP7, TP10, TP12, and TP16 are indicated.

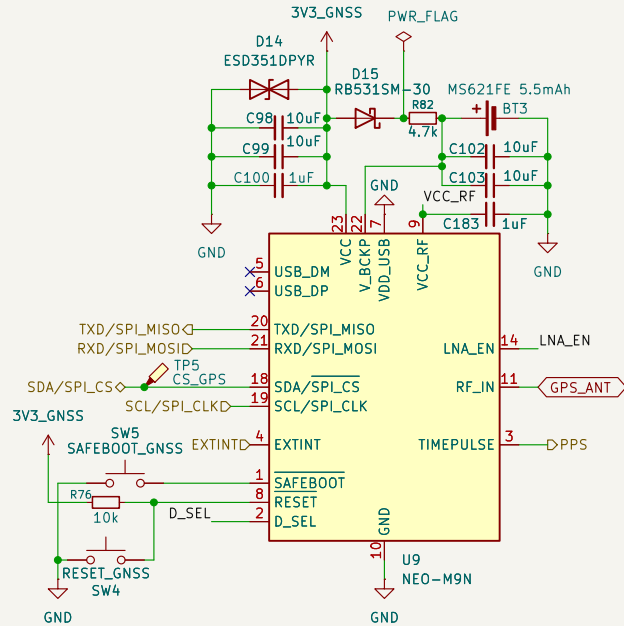
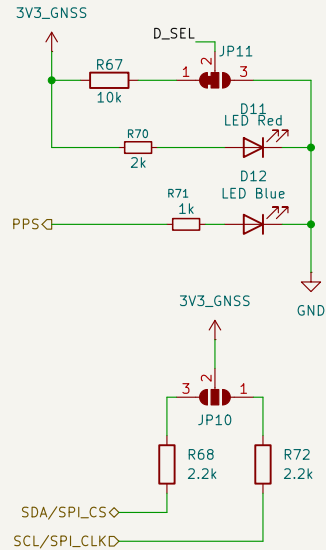


Title: Power Regulators

Rev:
Id: 4/7

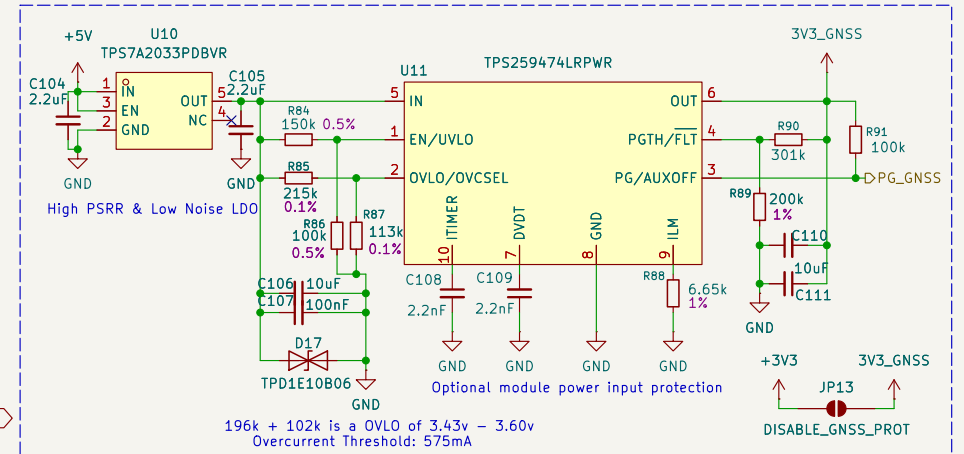
LEDs and Jumpers

High to enable I2C/UART, low to enable SPI



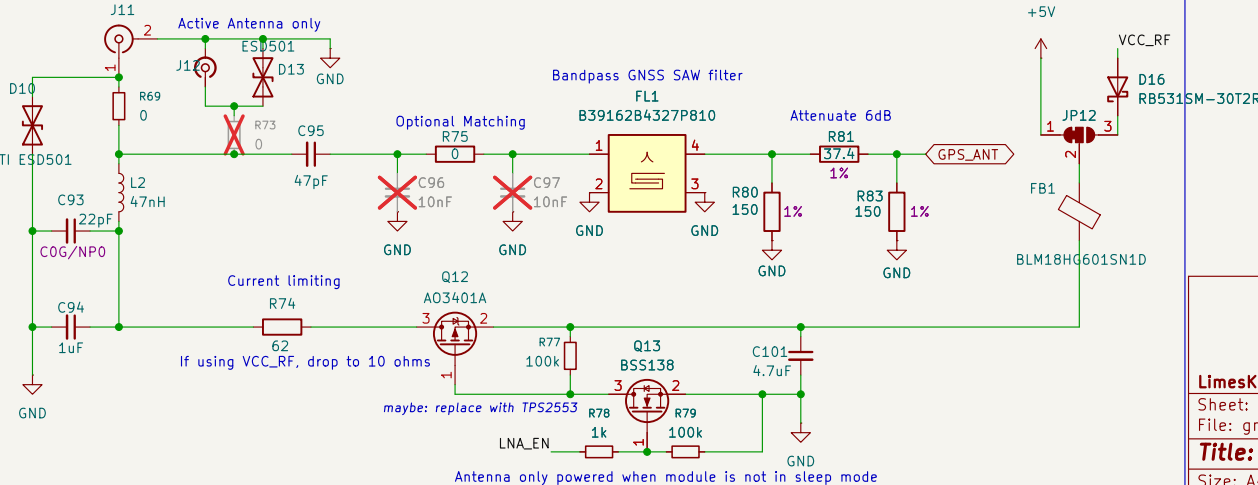
todo: switch module to use UART and not SPI
Theoretically compatible with NEO-M9N/M8N/M7N/M6N

Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf



196k + 102k is a OVLO of 3.43v - 3.60v
Overcurrent Threshold: 575mA

SMA + IPEX Connector



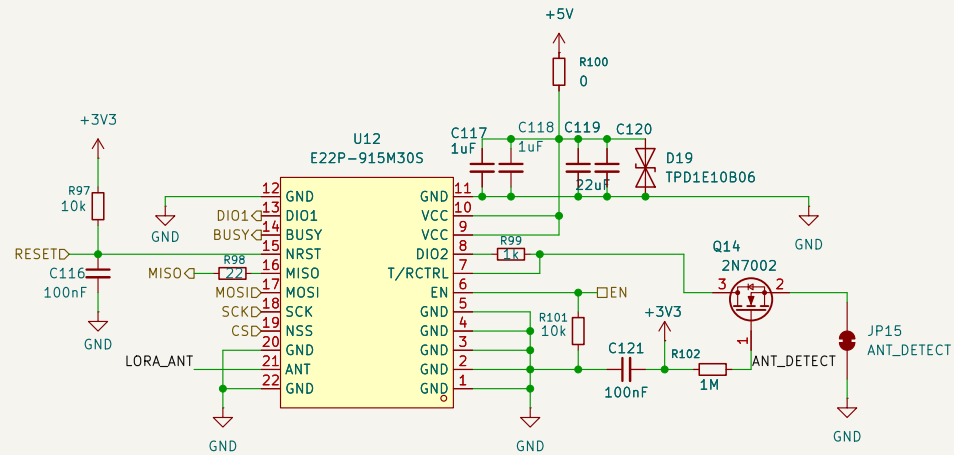
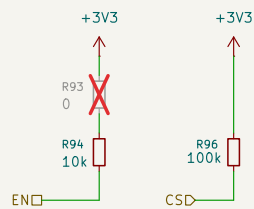
LimesKey

Sheet: /Root/GNSS/
File: gnss.kicad_sch

Title: GNSS Module / Active RF Antenna

Size: A4 Date: 2026-05-14
KiCad E.D.A. 10.0.6

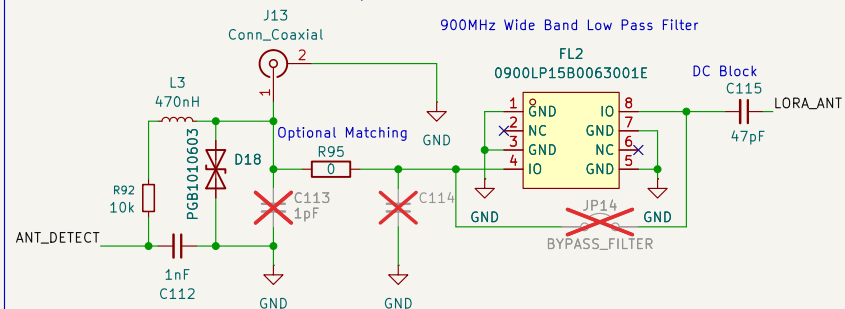
Rev:
Id: 5/7



Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4ffaf

SMA Connector

DC—shorted antenna only unless JP8 is cut



LimesKey

Sheet: /Root/LoRa/
 File: lora.kicad_sch

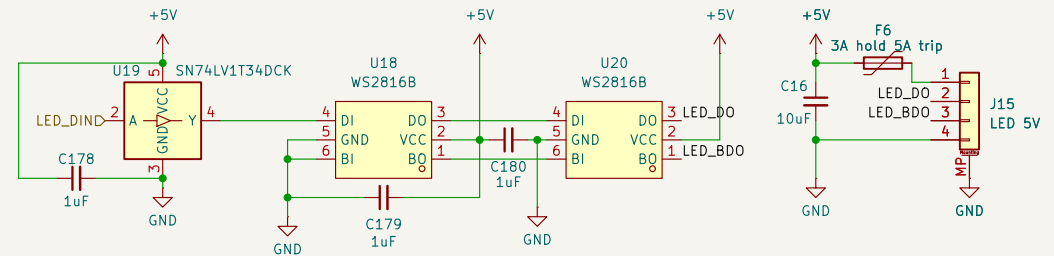
Title: LoRa Module / LoRa Antenna

Size: A4 Date: 2026-08-24
 KiCad E.D.A. 10.0.6

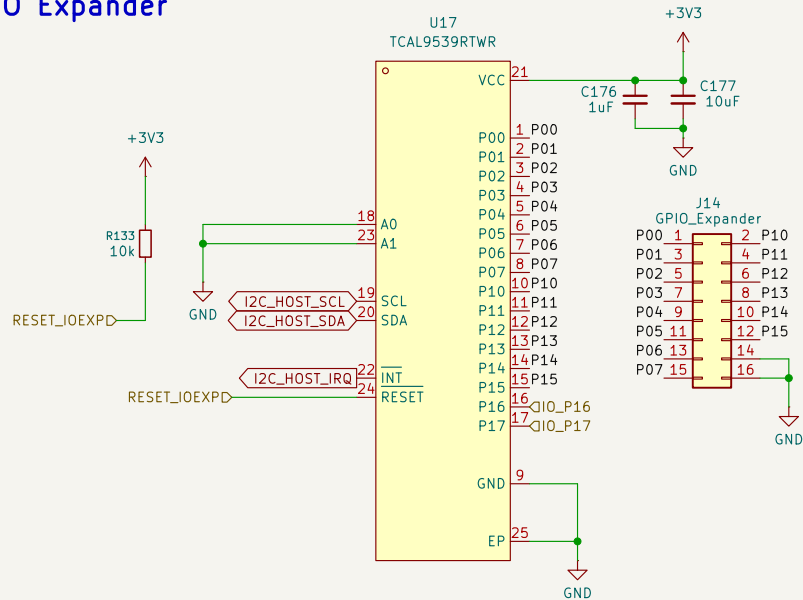
Rev:
 Id: 6/7

Individually Addressable RGB LEDs

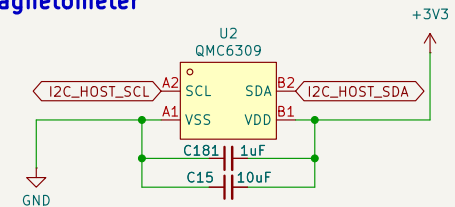
For use with WS2813/WS2816



GPIO Expander



Magnetometer



LimesKey

Sheet: /Root/Peripherals/
File: peripherals.kicad_sch

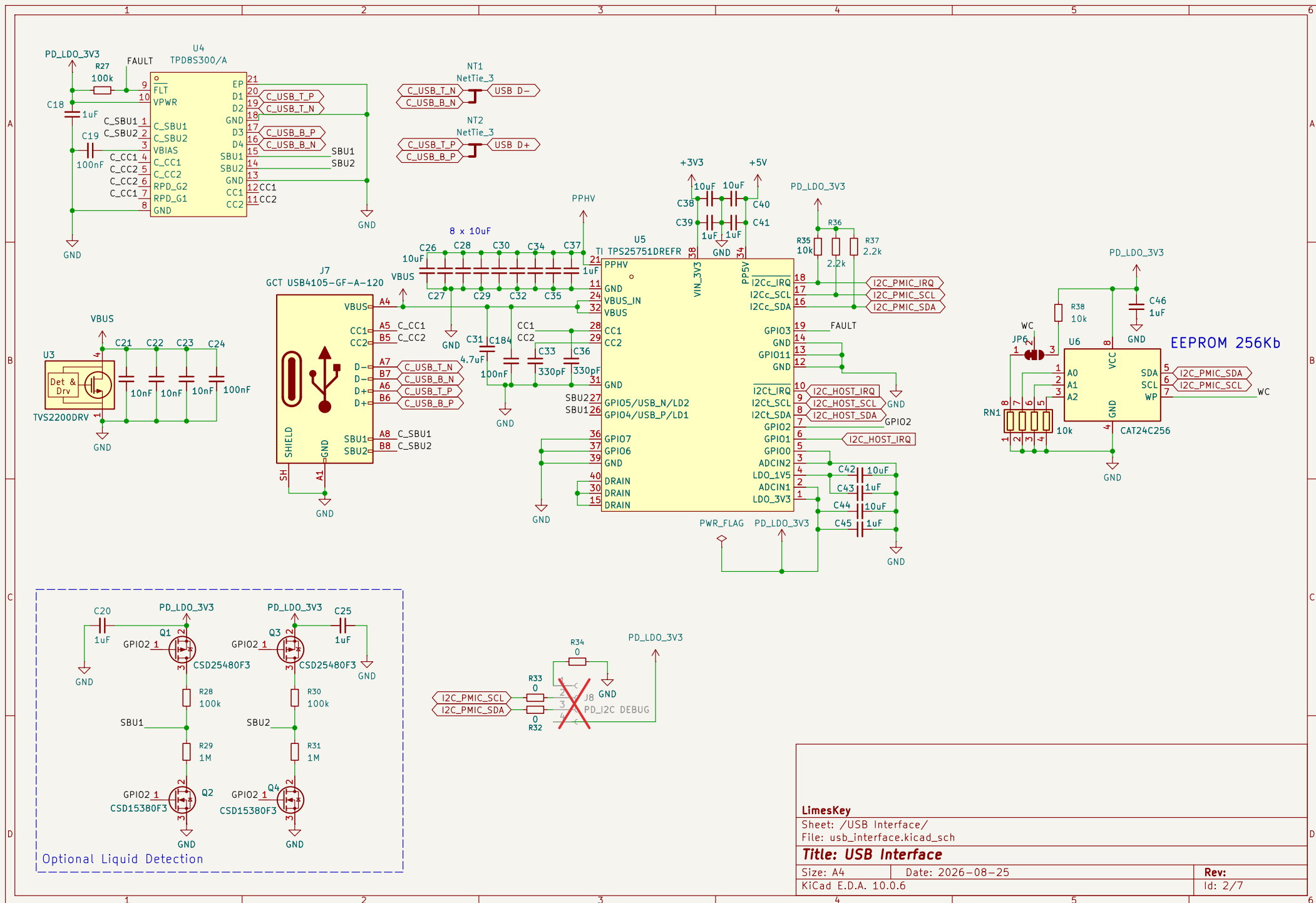
Title: GPIO Expander / LEDs / Magnetometer

Size: A4 Date: 2026-08-30

KiCad E.D.A. 10.0.6

Rev:

Id: 7/7



LimesKey

Sheet: /USB Interface/
File: usb_interface.kicad_sch

Title: USB Interface

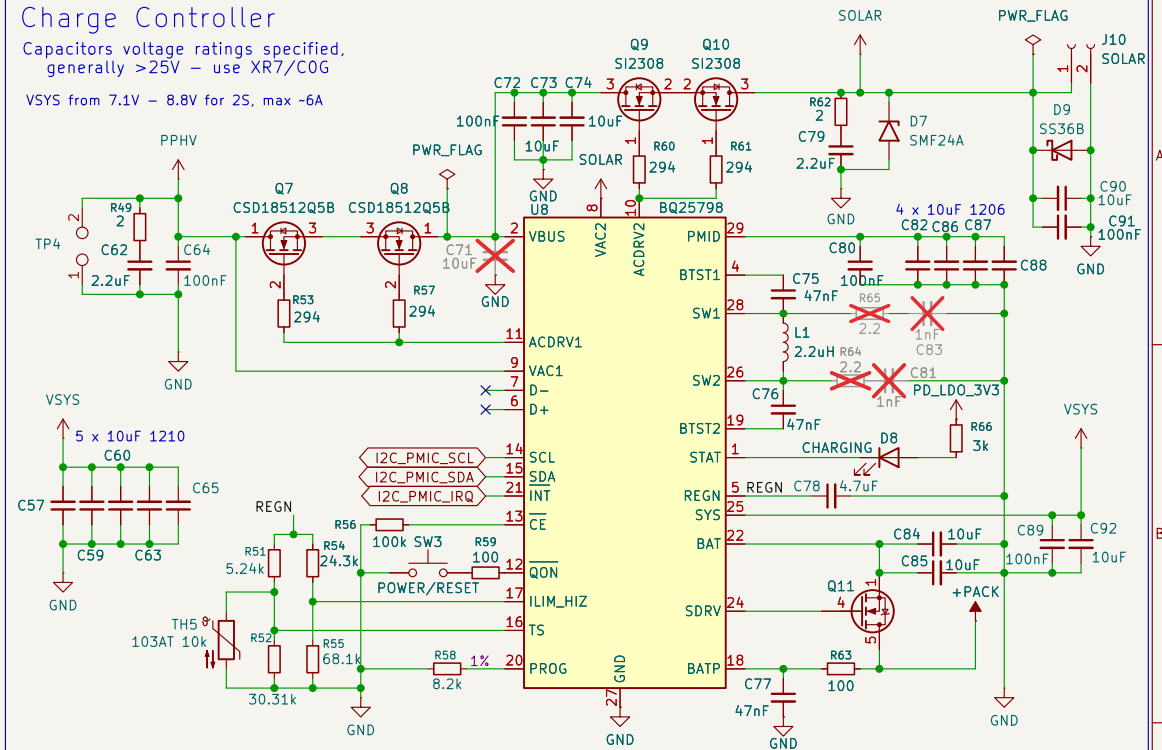
Size: A4 Date: 2026-08-25
KiCad E.D.A. 10.0.6

Rev:
Id: 2/7

Charge Controller

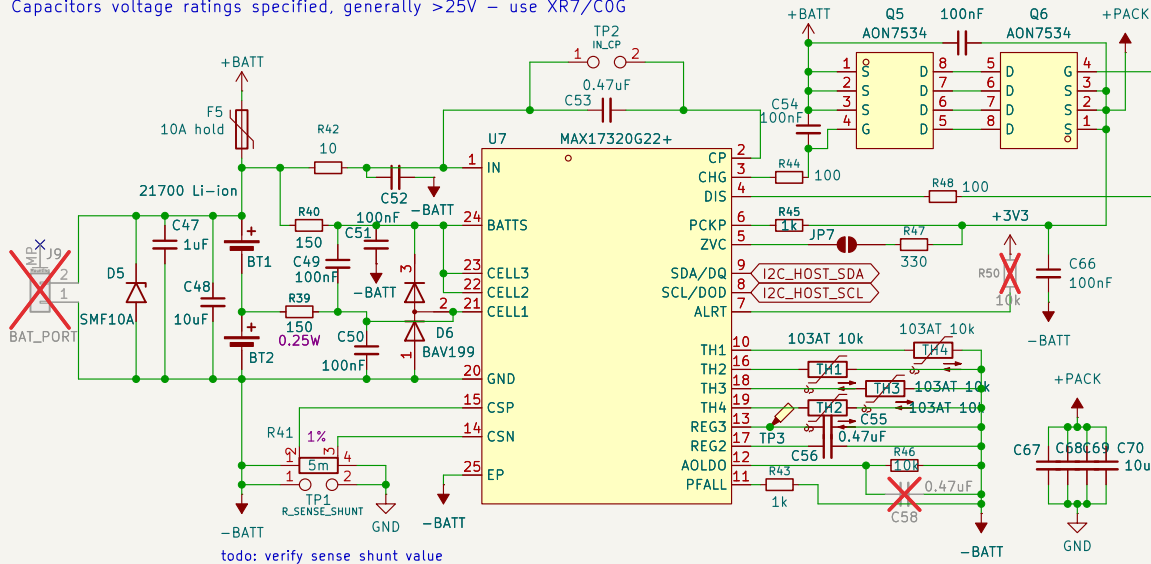
Capacitors voltage ratings specified,
generally >25V – use XR7/COG

VSYS from 7.1V – 8.8V for 2S, max -6A



Battery Management System (BMS)

Capacitors voltage ratings specified, generally >25V – use XR7/COG



LimesKey

Sheet: /Charger/
File: battery.kicad_sch

Title: Battery Charger / Battery Management System (BMS)

Size: A4 Date: 2026-08-24

KiCad E.D.A. 10.0.6

Rev:

Id: 3/7